



# Service Tool Instruction

Service Tool Instruction  
Number: 3377838-3

Revision Level:

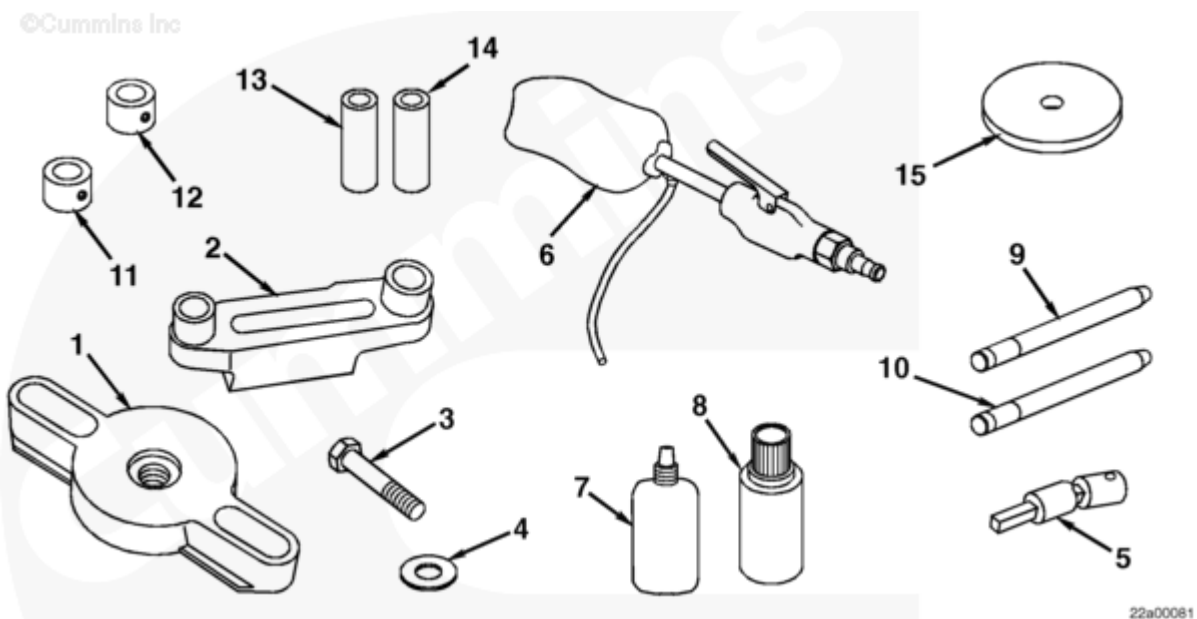
Released Date: 30-apr-  
2003

## Description

Thread Salvage Kit

## Purpose

This service tool instruction provides information for using the thread salvage kit, Part Number 3164021. The thread salvage kit is used in conjunction with the thread salvage inserts, drills, reamers, and taps listed on page 2 to salvage damaged cylinder head and main bearing capscrew threads in the engine block on several Cummins engine models.

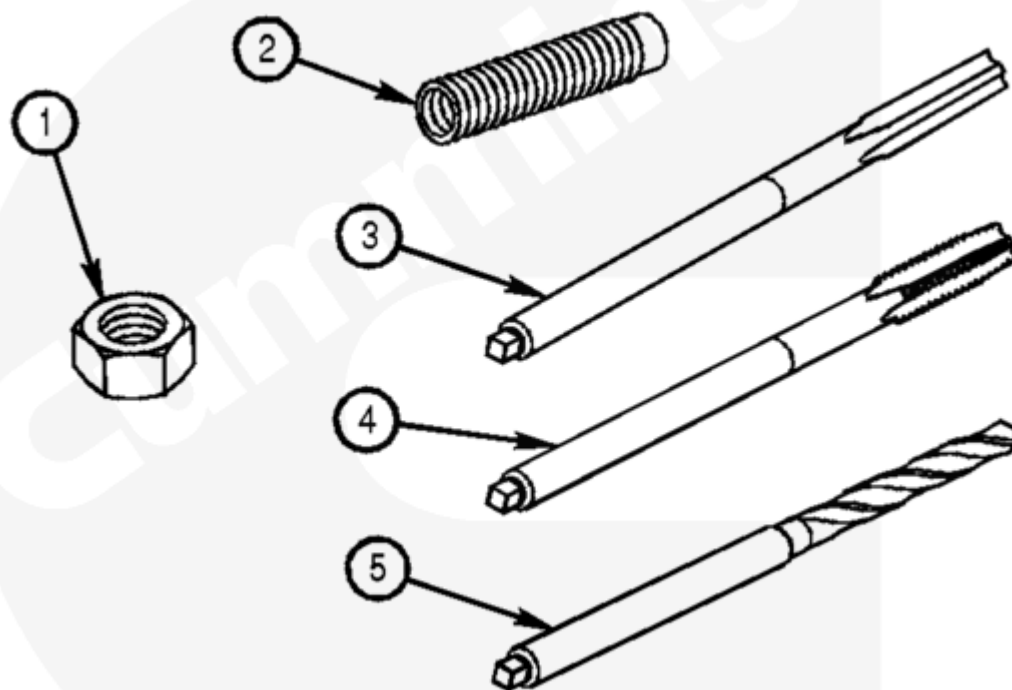


**Table 1, Thread Salvage Kit, Part Number 3164021**

| Item Number | Part Number | Description                           | Quantity |
|-------------|-------------|---------------------------------------|----------|
| 1           | 3164014     | Mounting plate                        | 1        |
| 2           | 3164015     | Drilling fixture                      | 1        |
| 3           | S-123B      | Hex head capscrew,<br>5/8-18 x 3-inch | 1        |
| 4           | S-658       | Heavy washer                          | 1        |
| 5           | 3164016     | Universal drive<br>socket             | 1        |

| Table 1, Thread Salvage Kit, Part Number 3164021 |             |                         |          |
|--------------------------------------------------|-------------|-------------------------|----------|
| Item Number                                      | Part Number | Description             | Quantity |
| 6                                                | 3164019     | Chip vacuum             | 1        |
| 7                                                | 3824038     | Thread-locking compound | 1        |
| 8                                                | 3824715     | Primer                  | 1        |
| 9                                                | 3377208     | Locating pin            | 1        |
| 10                                               | 3377209     | Locating pin            | 1        |
| 11                                               | 3164039     | Stop collar             | 1        |
| 12                                               | 3164020     | Stop collar             | 1        |
| 13                                               | 3164034     | Spacer                  | 2        |
| 14                                               | 3164035     | Spacer                  | 2        |
| 15                                               | 3376206     | Spacer                  | 1        |
| Not shown                                        | S-640       | Plain washer            | 4        |
| Not shown                                        | 3164240     | Storage case            | 1        |
| Not shown                                        | S-690       | Plain washer            | 4        |

| Table 2, Items Used with the Salvage Kit, Purchased Separately |             |                                |          |
|----------------------------------------------------------------|-------------|--------------------------------|----------|
| Item Number                                                    | Part Number | Description                    | Quantity |
| Not shown                                                      | 3823532     | Right-angle drill (heavy-duty) | 1        |



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**Table 3, Items Used to Salvage QST30 Cylinder Head Capscrew Threads, Purchased Separately**

| Item Number | Part Number | Description        | Quantity |
|-------------|-------------|--------------------|----------|
| 1           | 3251297     | Nut, jam (M16 x 2) | 1        |
| 2           | 3163563     | Insert, thread     | 1        |
| 3           | 3163898     | Reamer             | 1        |
| 4           | 3163900     | Tap                | 1        |
| 5           | 3164334     | Drill              | 1        |

**Table 4, Items Used to Salvage QSK45 and QSK60 Cylinder Head Capscrew Threads, Purchased Separately**

| Item Number | Part Number | Description          | Quantity |
|-------------|-------------|----------------------|----------|
| 1           | 3164241     | Nut, jam (M16 x 1.5) | 1        |
| 2           | 3163709     | Insert, thread       | 1        |
| 3           | 3163898     | Reamer               | 1        |
| 4           | 3163900     | Tap                  | 1        |
| 5           | 3164334     | Drill                | 1        |

**Table 5, Items Used to Salvage QSK78 Cylinder Head Capscrew Threads, Purchased Separately**

| Item Number | Part Number | Description          | Quantity |
|-------------|-------------|----------------------|----------|
| 1           | 3163841     | Nut, jam (M18 x 2.5) | 1        |
| 2           | 3163838     | Insert, thread       | 1        |
| 3           | 3163899     | Reamer               | 1        |
| 4           | 3163901     | Tap                  | 1        |
| 5           | 3164335     | Drill                | 1        |

**Table 6, Items Used to Salvage NH (5-1/8-in bore) Cylinder Head Capscrew Threads, Purchased Separately**

| Item Number | Part Number | Description       | Quantity |
|-------------|-------------|-------------------|----------|
| 1           | 3164343     | Nut, jam (7/8-14) | 1        |
| 2           | 3377133     | Insert, thread    | 1        |
| 3           | 3375202     | Reamer            | 1        |
| 4           | 3375203     | Tap               | 1        |

**Table 7, Items Used to Salvage NT (5-1/2-in bore), V903, and V28 Cylinder Head Capscrew Threads, Purchased Separately**

| Item Number | Part Number | Description                   | Quantity |
|-------------|-------------|-------------------------------|----------|
| 1           | 3164344     | Nut, jam (11/16-16)           | 1        |
| 2           | 3377134     | Insert, thread (short insert) | 1        |
| 2           | 3377135     | Insert, thread (long insert)  | 1        |
| 3           | 3376006     | Reamer                        | 1        |
| 4           | 3376007     | Tap                           | 1        |
| 5           | 3164334     | Drill                         | 1        |

**Table 8, Items Used to Salvage K19, K38, K50, and QSK19 Cylinder Head Capscrew Threads, Purchased Separately**

| Item Number | Part Number | Description                   | Quantity |
|-------------|-------------|-------------------------------|----------|
| 1           | S-249       | Nut, jam (3/4-16)             | 1        |
| 2           | 3377136     | Insert, thread (short insert) | 1        |
| 2           | 3377137     | Insert, thread (long insert)  | 1        |
| 3           | 3376204     | Reamer                        | 1        |
| 4           | 3376205     | Tap                           | 1        |
| 5           | 3164335     | Drill                         | 1        |

**Table 9, Items Used to Salvage QSK45 and QSK60 Main Bearing Capscrew Threads, Purchased Separately**

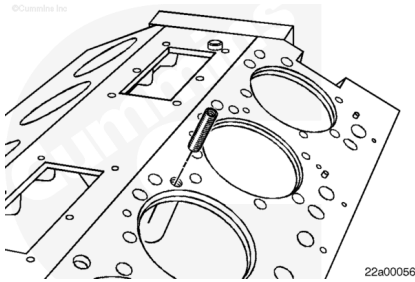
| Item Number | Part Number    | Description        | Quantity |
|-------------|----------------|--------------------|----------|
| 1           | 3165070        | Nut (M24-2)        | 1        |
| 2           | 3164961        | Insert, thread     | 1        |
| 3           | Not applicable | Reamer, 1.046-inch | 1        |
| 4           | Not applicable | Tap, 1.125-14      | 1        |
| 5           | Not applicable | Drill, 1-inch      | 1        |
| Not shown   | 3164964        | Spacer             | 1        |

**Table 10, Items Used to Salvage NH, NT, V28, K19, and K38 Main Bearing Capscrew Threads, Purchased Separately**

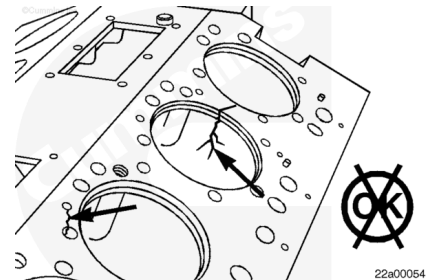
| Item Number | Part Number    | Description        | Quantity |
|-------------|----------------|--------------------|----------|
| 1           | Not applicable | Nut (1-14)         | 1        |
| 2           | 3376403        | Insert, thread     | 1        |
| 3           | Not applicable | Reamer, 1.046-inch | 1        |
| 4           | Not applicable | Tap, 1.125-14      | 1        |
| 5           | Not applicable | Drill, 1-inch      | 1        |

**Note :** This procedure is applicable for the engines listed in the tables, see pages 2 and 3.

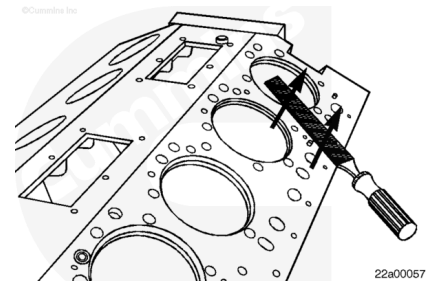
Follow this procedure to salvage capscrew holes that are missing a portion of threads or that have cracks extending into the threaded sidewall from the upper counterbore.



Cracks that extend into the cap screw bosses or the coolant cavity below the upper counterbore can **not** be salvaged using this procedure.



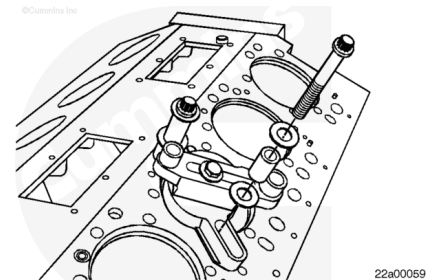
Remove any burrs from the top deck surface of the cylinder block. A 12-inch flat-surface mill file is effective for this step. Burr removal is necessary to obtain an accurate fixture location.



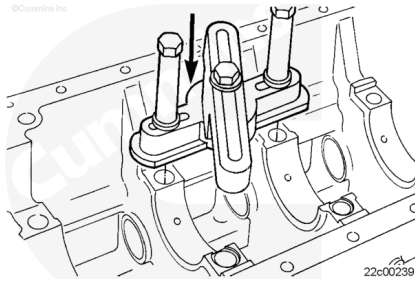
**Note :** The thread salvage kit is designed to be used with the cylinder liner removed. If the cylinder liner remains in the block, shims **must** be installed to both sides of the base equally to bridge the liner.

**Note :** On some engines, it will be necessary to use the spacer, Part Number 3376206, between the mounting base and drilling fixture.

Select two cap screw holes so the failed cap screw hole is positioned approximately in the middle. Install the mounting plate assembly onto the cap screw holes using the appropriate spacers, four plain washers, and the existing cylinder head cap screws. Use a heavy washer and the 5/8-18 x 3-inch hex head cap screw to attach the drilling fixture to the mounting plate assembly.



For main bearing cap screw thread repair, mounting plate, Part Number 3164014, will **not** accommodate all engines. In those cases, the repair **must** be performed by a machine shop.



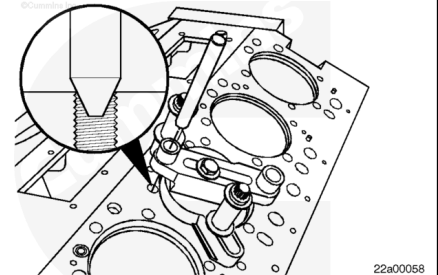
Slide the locating pin through the appropriate drilling fixture bushing hole until it rests in the failed capscrew hole of the cylinder block.

Center the drilling fixture guide bar and bushing over the failed capscrew hole.

Tighten the three capscrews to secure the fixture assembly.

**Torque Value:** 68 n·m [ 50 ft-lb ]

Remove the locating pin.

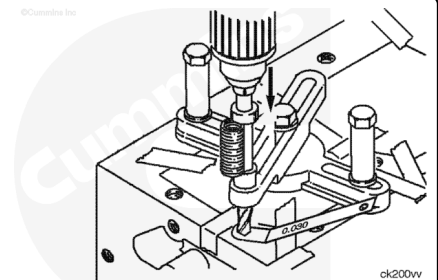


Setting Stop Collar (required **only** for repairs using drills)

Drills are used in some kits to allow the quick removal of waste material; however, care **must** be taken to make certain the depth of the repaired capscrew hole is **not** increased beyond specification.

The following procedure will result in the drill stopping short of the original hole depth. The remaining material is then removed with the reamer.

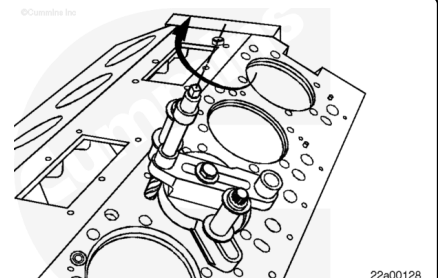
Place a 0.76-mm [0.030-in] feeler gauge over the hole to be repaired. Insert the drill through the drill bushing until its tip rests against the feeler gauge. Place the thread insert to be installed between the drill bushing and the stop collar. Move the stop collar down until it rests against the insert and tighten the stop collar.



**Note : This step is required only for repairs using drills.**

Place the drill through the drill fixture bushing and let it rest over the top of the failed capscrew hole.

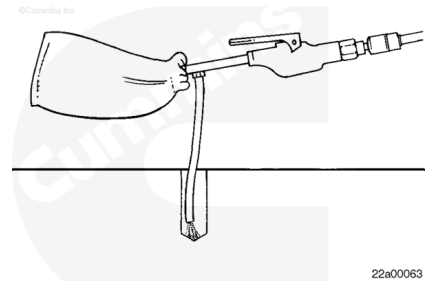
Use a drill motor with a clutch and a maximum operating speed of 300 rpm. Using a suitable cutting fluid, operate the drill motor until the stop collar contacts the drill bushing.



**⚠ WARNING ⚠**

**Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.**

Use the chip vacuum attached to compressed air to remove the chips from the newly-drilled hole.



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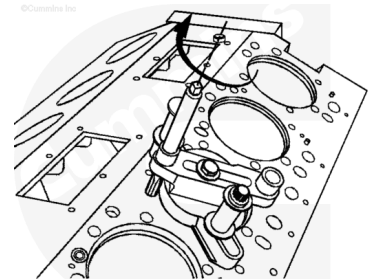
**Note : Some engines do not specify drills. In this case, all material is removed with a reamer.**

Place the reamer through the drill fixture bushing and let it rest over the top of the failed capscrew hole.

While reaming the hole, it will be necessary to stop frequently to clean the hole with the chip vacuum.

Use a drill motor with a clutch and a maximum operating speed of 300 rpm. Using a suitable cutting fluid, operate the drill motor until the reamer reaches the bottom of the original capscrew hole.

The reamer will stop cutting when it reaches the bottom of the hole.

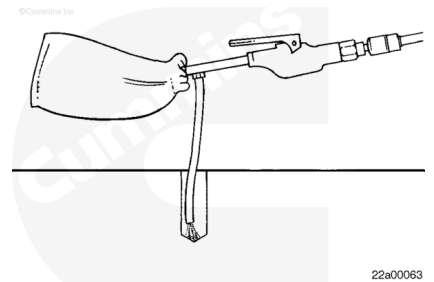


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**⚠ WARNING ⚠**

**Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.**

Use the chip vacuum attached to compressed air to remove the chips from the newly-reamed hole.



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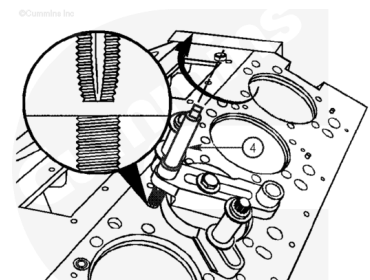
Stop frequently to clean the bore using a chip vacuum.

Install a 1/2-inch square drive socket on the end of the appropriate tap.

When tapping, keep the tap well lubricated.

Install the tap (4), and using a suitable tapping fluid, hand-tap to the bottom of the failed capscrew hole.

Remove the tap.



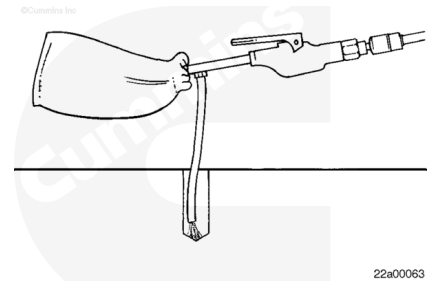
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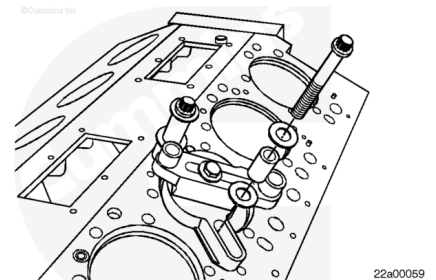
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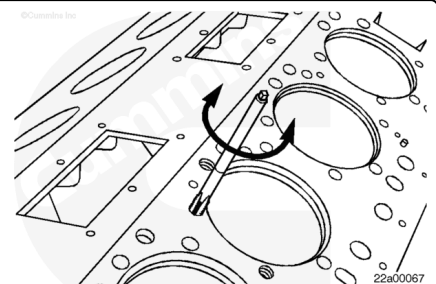
Use the chip vacuum attached to compressed air to remove the chips from the newly-tapped hole.



Remove the mounting capscrews, spacers, four plain washers, and mounting plate assembly.

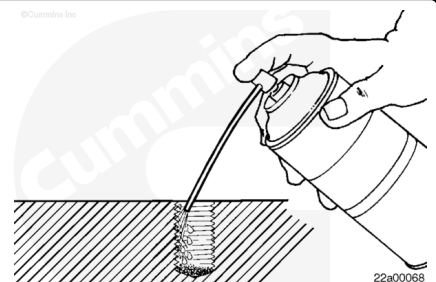


Thread the appropriate tap in and out of the newly-cut threads several times to verify the thread condition and clean debris from the threads.



**⚠ WARNING ⚠**

**When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.**



**⚠ WARNING ⚠**

**Some solvents are flammable and toxic. Read the manufacturer's instructions before using.**

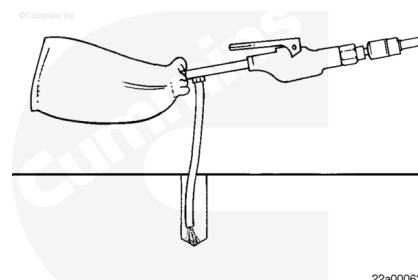
Check the capscrew hole for evidence of porosity and cracks.

Clean the threads and flush any debris from the tapped hole using a solvent-degreasing agent.

**⚠ WARNING ⚠**

**Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.**

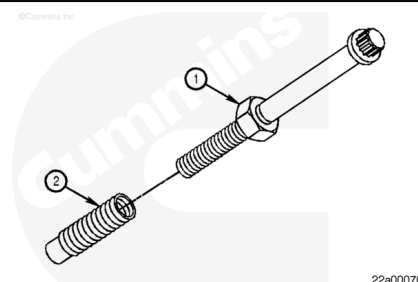
Use the chip vacuum attached to compressed air to remove the solvent-degreasing agent from the newly-tapped hole.



Place the appropriate jam nut (1) on an appropriate capscrew.

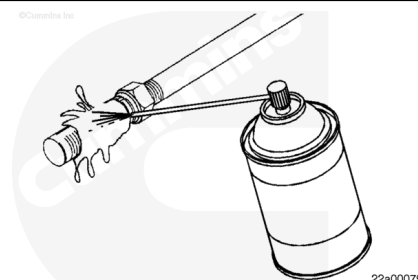
Install the appropriate thread insert (2) onto the cylinder head mounting capscrew until it nears the bottom of the insert.

Rotate the jam nut (1) until it contacts the thread insert (2) to lock the insert and jam nut together.



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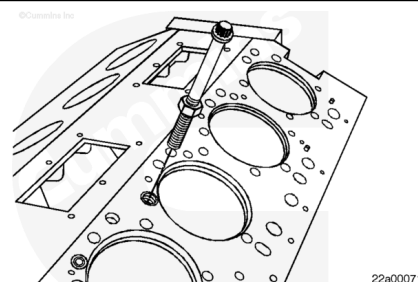
**⚠ WARNING ⚠**

**Some solvents are flammable and toxic. Read the manufacturer's instructions before using.**

Remove the preservative coating from the circumference of the thread insert using a solvent-degreasing agent.

**Note :** For some applications, it is necessary to install a spacer between the thread insert and the jam nut. Refer to the tables on pages 2 and 3 for spacer information.

Install the thread insert, jam nut, and capscrew into the newly-tapped hole. Make certain that the thread insert threads freely into the hole.



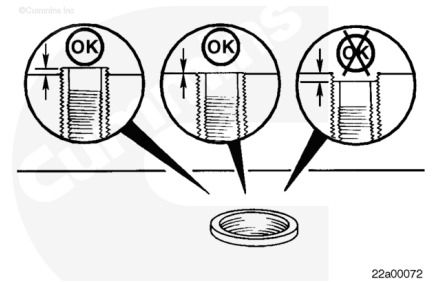
**Note :** For applications with a counterbore, measure the protrusion depth starting at the counterbore face, which is recessed into the hole.

Check for protrusion. Make certain the thread insert protrusion is 0 to 1.0 mm [0 to 0.040 in] above the cylinder block surface.

It is important to make certain the minimum required thread depth is maintained after thread repair.

If the insert protrusion exceeds 1 mm [0.040 in], the cylinder head capscrew can bottom out when tightened. In this case, it will be necessary to increase the depth of the original capscrew hole.

If the insert protrusion is correct, proceed to the Install Insert procedure. If the protrusion is **not** correct, record the measurement for use in the Adjust Bore Depth procedure.

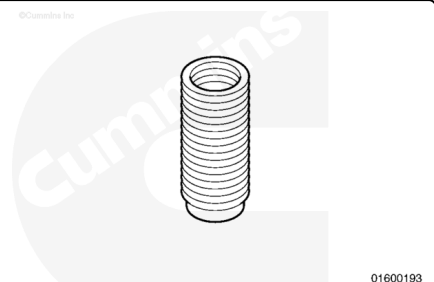


Prior to increasing the capscrew hole depth, check the following:

1. Make certain the correct thread insert is being used. Some engines use more than one insert; typically, these inserts will be identical except for their length. Make certain the insert used is the correct length.
2. Make certain the capscrew hole is threaded all the way to the bottom.
3. Make certain there are no burrs or other damage that would prevent the insert from threading to the bottom of the capscrew hole.

After checking the above listed items, if the thread insert protrusion still exceeds 1 mm [0.040 in], the depth of the original capscrew hole depth **must** be increased.

Proceed to the next procedure.



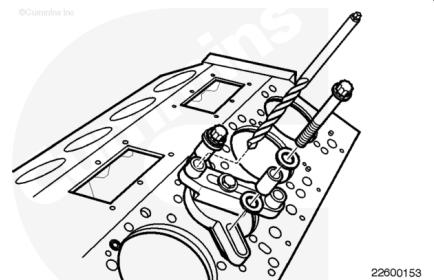
**Note :** The thread salvage kit is designed to be used with the cylinder liner removed. If the cylinder liner remains in the block, shims **must** be installed to both sides of the base equally to bridge the liner.

**Note :** On some engines, it will be necessary to use the spacer, Part Number 3376206, between the mounting base and drilling fixture.

**Note :** This procedure can **only** be performed if the correct drill is available.

Reinstall the mounting plate assembly. Leave all fasteners loose.

Slide the drill used previously through the drill bushing and into the threaded hole until it rests at the bottom of the hole.



Tighten all fasteners.

**Torque Value:** 68 n•m [ 50 ft-lb ]

The drill **must** slide freely in the bushing and threaded hole, indicating that it is centered in the hole.

Do **not** increase the depth of the original capscrew hole more than 3 mm [0.120 in].

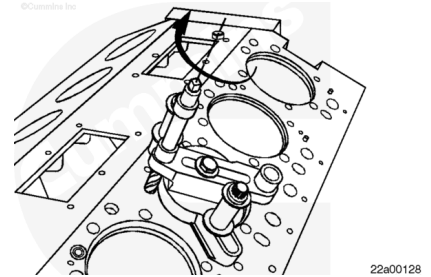
Slide the stop collar over the drill bit.

Subtract 0.50 mm [0.020 in] from the insert protrusion recorded in the Insert Protrusion procedure.

Select a feeler gauge equivalent to this value.

With the drill resting at the bottom of the hole, insert the feeler gauge between the stop collar and drill bushing. Tighten the stop collar.

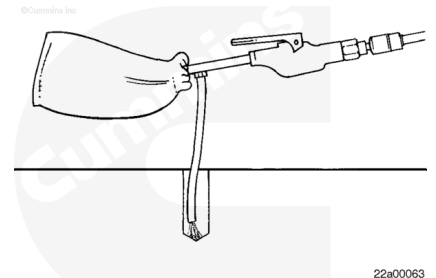
Use a drill motor with a clutch and a maximum operating speed of 300 rpm. Using a suitable cutting fluid, operate the drill motor until the stop collar contacts the drill bushing.



**⚠ WARNING ⚠**

**Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.**

Use the chip vacuum attached to compressed air to remove the chips from the newly-drilled hole.



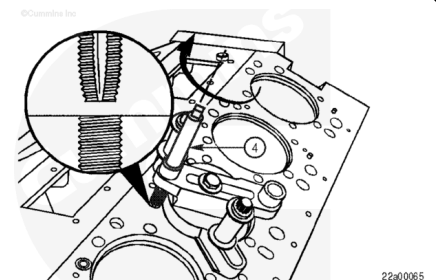
It is **not** necessary to ream the hole prior to re-tapping.

Install a 1/2-inch square drive socket on the end of the appropriate tap.

When tapping, keep the tap well lubricated.

Install the tap (4), and using a suitable tapping fluid, hand-tap to the bottom of the failed capscrew hole.

Remove the tap.



**⚠ WARNING ⚠**

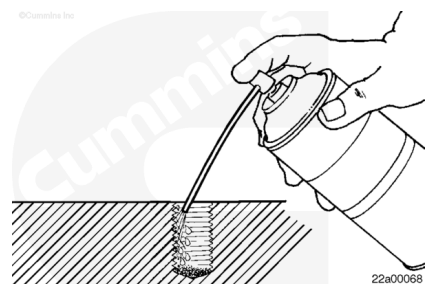
When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.

**⚠ WARNING ⚠**

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Check the capscrew hole for evidence of porosity and cracks.

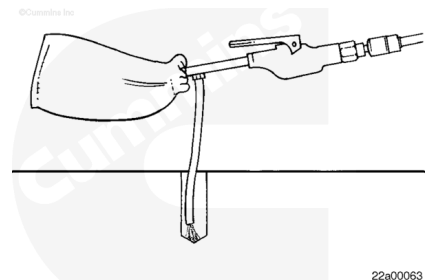
Clean the threads, and flush any debris from the tapped hole using a solvent-degreasing agent.



**⚠ WARNING ⚠**

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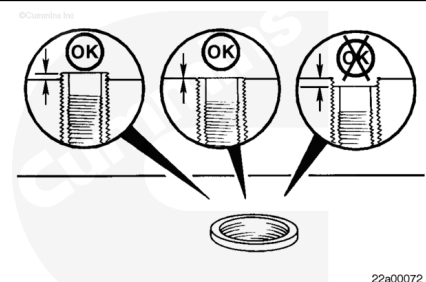
Use the chip vacuum attached to compressed air to remove the chips and solvent-degreasing agent from the tapped hole.



Install the thread insert on a cylinder head capscrew and lock in place with a jam nut.

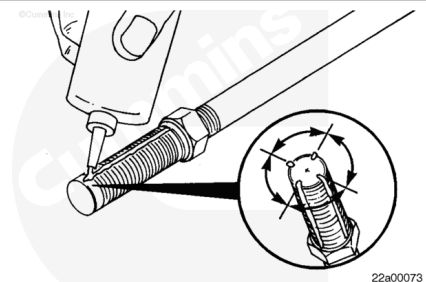
Temporarily install the thread insert and check/recheck the protrusion. Make certain that the protrusion is 0 to 1 mm [0 to 0.040 in] above the cylinder block surface.

Remove the insert.



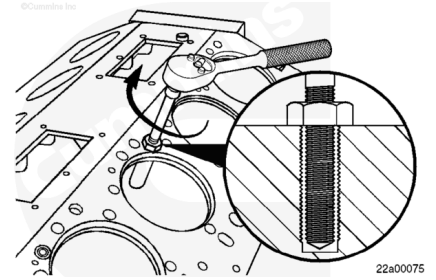
Apply a light coat of primer, Part Number 3824715, to the thread insert and the threaded hole. Allow three to five minutes to dry.

Apply four beads of thread locking compound, Part Number 3824038, to the circumference of the thread insert. Each bead **must** be 0.8 mm [0.03 in] wide down the entire length of the thread insert and the beads **must** be 90 degrees apart.



Install the thread insert until it is flush with the cylinder block surface or it bottoms in the hole.

Allow to dry for three hours.



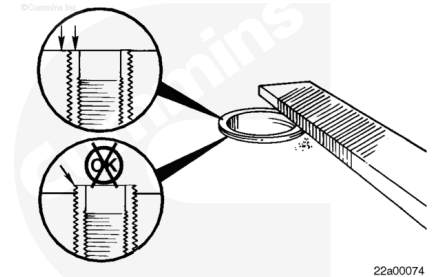
**⚠ WARNING ⚠**

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Remove the capscrew and jam nut.

If necessary, file or machine the top of the thread insert so it is flush with the cylinder block surface.

Make certain that the inner diameter of the thread insert is free of burrs. Remove any file shavings or debris with the chip vacuum.



For applications where the thread insert is recessed into a counterbore, an end mill or a machine cutter will be necessary to remove the excess length of the insert and restore the alignment of counterbore.

